

Recurrence for an arbitrary DAG

let $s(b)$ be the length of a longest path from source to b

$$s_b = \max_{\substack{\text{all predecessors } a \\ \text{of node } b}} \{s_a + \text{weight of edge from } a \text{ to } b\}$$

For a given node, you need to process all of its predecessors

Topological Ordering

an ordering of nodes in a DAG s.t. every edge $a_i \rightarrow a_j$ has $i < j$

This algorithm is $O(V+E)$ (loop through each node and each edge of that node).

Amount of edges in Alignment graph is roughly $n \cdot m$

To recover the original path just store which one you took

Might assign different penalties to mismatch and in/del
 μ σ

We just have to change the weights

BLISSUM Scoring Matrices $\rightarrow$ Done wrong and realized after 15 years

(Why we weigh things differently)

Global Alignment Problem (We just lookup edge weights in the matrix).

Protein-level functions are more informative (bigger alphabet etc.)